

Energy Efficiency Optimization in IRS-Assisted Multi-UAV Networks using a DRL Approach

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Abstract—In this paper, we consider the deployment of multiple unmanned aerial vehicles (UAVs), each equipped with an intelligent reflecting surface (IRS) to serve ground users in highly dynamic and challenging environments (e.g., disaster areas, emergency zones). To begin with, we investigate the joint optimization of the IRS phase shift and UAV trajectory using a soft actor critic (SAC) approach to maximize the energy efficiency in the proposed IRS-assisted multi-UAV network. In addition, we compare the proposed SAC-based framework with existing machine learning methods in terms of energy efficiency and demonstrate its superiority over the state of the art through numerical results. Finally, we quantify the percentage gain from each optimization design (i.e., UAV trajectory and IRS phase shift) that contributes to the overall improvement of the energy efficiency.

Index Terms—Deep reinforcement learning, energy efficiency, intelligent reflecting surface, unmanned aerial vehicle.

I. INTRODUCTION

FUTURE wireless networks are expected to integrate terrestrial and aerial communications, paving the way for better coverage and improved connectivity, particularly in disaster-stricken zones or complex urban environments. To this end, unmanned aerial vehicle (UAV) technology has recently emerged as a key enabler due to its versatility, flexibility, and cost-effectiveness. Despite these attractive features, a standalone UAV technology may not be sufficient to address coverage issues, particularly in scenarios where communications among terrestrial users are non-existent as a result of natural disasters or the inherent nature of challenging environments. In this regard, integrating UAV with a new emerging technology such as intelligent reflecting surface (IRS) can create a robust framework. An IRS is a smart reflective surface consisting of a large number of programmable metamaterial elements that can be dynamically controlled to manipulate the propagation environment [1].

Recent studies have explored the synergy between UAV and IRS technologies to overcome terrestrial obstacles, thus improving reliability, energy efficiency, and signal coverage (see [2]–[4] and references therein). This has further motivated performance optimization (e.g., data rates and energy efficiency) of IRS-aided UAV systems. In [4], the authors investigated an IRS-assisted UAV communication system with the aim of maximizing the achievable rate. To this end, a beamforming and trajectory optimization problem was formulated and solved by using a successive convex approximation (SCA) method. Using double deep Q-network (DDQN) in [5], the authors studied a dual-objective optimization problem that

maximizes the data rate while minimizing the UAV energy propulsion with a ground-based IRS. In [6], Chen *et al.* studied a novel UAV decode-and-forward (DF) relay network with multiple IRSs by formulating a joint optimization problem that involves sub-channel allocation and multi-IRS phase shifts. Unlike [4]–[6], the work of [7] considered a UAV-mounted IRS configuration to provide greater flexibility and versatility. Given this setup, Truong *et al.* investigated a UAV-mounted IRS downlink multiple-input single-output (MISO) system and used a deep reinforcement learning (DRL) framework based on the deep deterministic policy gradient (DDPG) algorithm to solve a joint optimization problem involving the UAV trajectory and IRS phase shift, which maximizes the achievable system sum rate.

Common to the aforementioned studies, energy efficiency (EE), which is a critical metric for the next-generation of wireless technology, has not been considered. In [8], a joint optimization problem that maximizes the EE rate of a UAV network was solved using an adaptive deep reinforcement learning (ADRL) approach. However, no IRS was considered. Despite recent advances in aerial IRS-based wireless networks and to the best of our knowledge, very few studies have focused on maximizing the EE rate as the key performance metric. This research gap is critical, as maximization of EE is essential for the development of practical UAV-mounted IRS networks. Moreover, it remains an open question how much the UAV trajectory and IRS phase shift individually contribute to the overall performance improvement. Motivated by these observations, we investigate a joint UAV trajectory and IRS phase shift optimization that maximizes the EE rate of the proposed UAV-mounted IRS network.

The contributions of the proposed work are summarized as follows. First, unlike traditional optimization methods (e.g., SCA, block coordinate descent (BCD)) widely explored in the existing literature that cannot adapt to dynamic environments, we investigate a DRL approach in the soft actor-critic (SAC) for the joint optimization of the IRS trajectory and phase shift of the proposed IRS-assisted UAV network. Second, we quantify the contribution of each component of the joint optimization (i.e., IRS phase shift and UAV trajectory) to the overall performance gain. This decomposition may help prioritize optimization efforts to yield maximum EE gains in future UAV-IRS designs. Third, we compare the proposed SAC algorithm with some existing state-of-the-art machine learning (ML) approaches based on application requirements.

Finally, we provide some numerical results to assess the effectiveness of the proposed system. Table I summarizes the comparison between our contributions with some state-of-the-art ML methods on IRS-assisted UAV.

TABLE I: COMPARISON OF THE PROPOSED FRAMEWORK WITH STATE-OF-THE-ART ML RELATED STUDIES.

Work	Algorithm	UAV-IRS	EE Focus	Decomp.
[4]	SCA	Single	No	No
[5]	DDQN	Ground	Partial	No
[6]	SAC-PER	Ground	No	No
[7]	DDPG	Single	No	No
[8]	ADRL	No IRS	Yes	Partial
This work	SAC	Multiple	Yes	Yes

The remainder of the paper is structured as follows. The system is described in Section II followed by the proposed framework in Section III. The numerical results are discussed in Section IV, and some concluding remarks are provided in Section V.

II. SYSTEM MODEL

We consider a disaster communication scenario where M UAVs, each equipped with an IRS comprising N passive reflecting elements, serve K ground users within an area $\mathcal{A}$ of dimension 1000×1000 m². The system operates in discrete time slots $t \in \{1, 2, \dots, T\}$, where a single base station (BS) communicates with ground users through UAV-mounted IRS links. This multi-UAV architecture differs from prior single-UAV studies [4], [7] by enabling coordinated coverage and load balancing across the disaster area.

A. UAV Mobility Model

Each UAV m maintains a three-dimensional position $\mathbf{q}_m(t) = [x_m(t), y_m(t), h_m(t)]^T$, constrained within altitude bounds $h_{\min} \leq h_m(t) \leq h_{\max}$, where $h_{\min}$ and $h_{\max}$ are the minimum and maximum allowable heights at which a UAV can fly, respectively. In addition, the velocity is bounded by $\|\mathbf{v}_m(t)\| \leq v_{\max}$, where $v_{\max}$ is the maximum velocity of a UAV, and position updates follow $\mathbf{q}_m(t+1) = \mathbf{q}_m(t) + \mathbf{v}_m(t)\delta_t$, with δ_t denoting the duration of a time slot. A minimum inter-UAV separation of $d_{\min}$ is enforced to prevent collisions while enabling coordinated swarm operation.

B. Channel Model

The channel between BS and the m -th UAV channel is assumed to follow a Rician distribution with the corresponding channel coefficient given by

$$\mathbf{h}_{\text{BU},m} = \sqrt{\frac{\kappa_r}{\kappa_r + 1}} \mathbf{h}_{\text{LoS}} + \sqrt{\frac{1}{\kappa_r + 1}} \mathbf{h}_{\text{NLoS}}, \quad (1)$$

where κ_r is the K-factor capturing the strong line-of-sight (LoS) component typical of aerial links, $\mathbf{h}_{\text{LoS}} = e^{-j2\pi d_m/\lambda} \mathbf{a}(\phi_m)$ represents the deterministic LoS component with steering vector $\mathbf{a}(\phi_m)$, and $\mathbf{h}_{\text{NLoS}} \sim \mathcal{CN}(\mathbf{0}, \mathbf{I})$ captures the multipath scattering. Moreover, the UAV-to-user channels experience distance-dependent path loss with exponent α , following the probabilistic LoS model for urban environments [9]. Thus, the effective channel through the m -th IRS is expressed as

$$h_{\text{eff},m,k}(t) = \mathbf{h}_{\text{UK},m}^H \Phi_m(t) \mathbf{h}_{\text{BU},m}, \quad (2)$$

where $\Phi_m(t) = \text{diag}(e^{j\theta_{m,1}(t)}, \dots, e^{j\theta_{m,N}(t)})$ represents the IRS phase shift matrix with $\theta_{m,n}(t) \in [0, 2\pi]$, and $\mathbf{h}_{\text{UK},m}$ denotes the channel from the m -th IRS to user k . According to [10], it follows that the optimal phase alignment provides 12–15 dB passive beamforming gain, effectively multiplying signal power without active amplification.

C. Energy Consumption Model

In what follows, we examine the total system power which is given by

$$P_{\text{total}}(t) = \sum_{m=1}^M [P_{\text{hover}} + P_{\text{move},m}(t) + P_{\text{IRS},m} + P_{\text{proc}}], \quad (3)$$

where P_{hover} represents hovering power of the rotary-wing UAVs, $P_{\text{move},m}(t) = \kappa \|\mathbf{v}_m(t)\|$ accounts for velocity-dependent propulsion, $P_{\text{IRS},m} = \eta N$ with η denoting the IRS circuit power, and P_{proc} is the processing overhead. This model captures the propulsion-dominated nature of UAV energy consumption, where flight mechanics consume over 95% of total power [11].

D. Problem Formulation

In this subsection, we formulate a joint optimization problem that maximizes the EE based on the system achievable rate. To begin with, we define the achievable rate for user k , which is given by

$$R_k(t) = B \log_2 \left(1 + \frac{|h_{\text{eff},k}(t)|^2 P_{\text{tx}}}{\sigma^2} \right), \quad (4)$$

where B , P_{tx} , and σ^2 denote the bandwidth, transmit power, and noise power, respectively. Hence, the joint optimization problem to maximize EE denoted by η_{EE} can be formulated as

$$\max_{\{\mathbf{q}_m(t), \theta_m(t)\}} \eta_{\text{EE}} = \frac{\sum_{k=1}^K R_k(t)}{P_{\text{total}}(t)} \quad (5a)$$

$$\text{s.t. } R_k(t) \geq R_{\min}, \quad \forall k, \quad (5b)$$

$$\|\mathbf{v}_m(t)\| \leq v_{\max}, \quad \forall m, \quad (5c)$$

$$h_{\min} \leq h_m(t) \leq h_{\max}, \quad \forall m, \quad (5d)$$

$$\|\mathbf{q}_m(t) - \mathbf{q}_{m'}(t)\| \geq d_{\min}, \quad \forall m \neq m', \quad (5e)$$

$$\theta_{m,n}(t) \in [0, 2\pi], \quad \forall m, n. \quad (5f)$$

The aforementioned formulated problem is non-convex and NP-hard due to the coupled optimization variables $\mathbf{q}_m(t)$ and $\theta_m(t)$, which appear to be multiplicative in the effective channel expression of (2). Traditional approaches such as SCA [4] and BCD [10] require 150–200 iterations with $\mathcal{O}(N^3 M)$ complexity, rendering them unsuitable for real-time control in dynamic environments with a channel coherence time of 100 ms.

III. PROPOSED FRAMEWORK

To address the non-convex optimization problem in (5), we propose a SAC-based DRL framework that learns a joint trajectory and phase shift control policy through entropy-regularized reinforcement learning (RL). The environment state feeds into an SAC agent, which outputs continuous

actions for trajectory and phase control. The performance metrics drive reward calculation through a feedback loop. Unlike DDPG employed in [7] which uses deterministic policies susceptible to local optima, SAC employs stochastic policies with maximum entropy exploration, thus enabling a robust optimization in high-dimensional action spaces. Fig. 1 illustrates the proposed framework architecture.

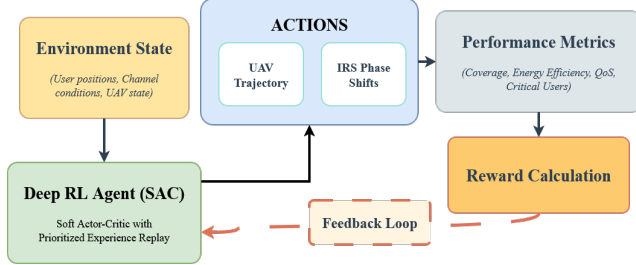


Fig. 1: Proposed SAC framework for joint UAV trajectory and IRS phase shift optimization.

A. Markov Decision Process Formulation

The optimization problem is formulated as a Markov decision process (MDP) defined by the tuple $(\mathcal{S}, \mathcal{A}, \mathcal{R}, \gamma)$.

State Space: The state $s(t) \in \mathcal{S}$ encompasses the channel state information (CSI), UAV positions, current phase configurations, battery levels, and user metrics. The state dimension scales as $\mathcal{O}(MN + 3M + K)$, yielding a 148-dimensional state vector for our configuration.

Action Space: The action $\mathbf{a}(t) \in \mathcal{A}$ comprises continuous velocity commands $\mathbf{v}_m(t) \in [-v_{\max}, v_{\max}]^3$ for each UAV and phase shift updates $\theta_m(t) \in [0, 2\pi]^N$ for each IRS, yielding an action dimension of $3M + MN = 70$ for joint trajectory-phase control.

Reward Function: The multi-objective reward balances EE, coverage, and constraint satisfaction as

$$r(t) = w_c r_c + w_{\text{crit}} r_{\text{crit}} + w_{\text{ee}} r_{\text{ee}} + w_{\text{qos}} r_{\text{qos}} - \mathcal{P}(t), \quad (6)$$

where r_c denotes the coverage reward with super-linear penalties below 50% coverage, r_{crit} prioritises critical users in disaster scenarios, r_{ee} represents the EE reward, r_{qos} ensures quality-of-service, and $\mathcal{P}(t)$ penalizes battery depletion, collisions, and boundary violations. The weights are set as $w_c = 0.40$, $w_{\text{crit}} = 0.30$, $w_{\text{ee}} = 0.15$, and $w_{\text{qos}} = 0.10$. This weighting prioritizes coverage (the primary disaster response objective) whilst maintaining energy awareness; alternative weightings would shift the Pareto front position in Fig. 2.

B. Soft Actor-Critic Algorithm

SAC provides three characteristics relevant to the UAV-IRS optimization: (i) continuous action space handling for phase shifts $\theta \in [0, 2\pi]$ and velocities without discretization losses incurred by deep Q-network (DQN) [12]; (ii) off-policy learning with experience replay for sample efficiency compared to on-policy methods such as proximal policy optimisation (PPO) [13]; and (iii) maximum entropy regularization preventing premature convergence in the 70-dimensional action space. The SAC objective maximizes the expected return augmented with entropy:

$$J(\pi) = \mathbb{E}_{\tau \sim \pi} \left[\sum_{t=0}^T \gamma^t (r(s_t, \mathbf{a}_t) + \alpha^* \mathcal{H}(\pi(\cdot|s_t))) \right], \quad (7)$$

where γ is the discount factor, α^* is the temperature parameter with automatic tuning for adaptive exploration, and $\mathcal{H}(\pi) = -\mathbb{E}[\log \pi]$ denotes the policy entropy.

The actor network $\pi_\phi(\mathbf{a}|s)$ outputs a Gaussian distribution with mean $\mu(s)$ and standard deviation $\sigma(s)$, enabling stochastic policy exploration across the non-convex landscape. Twin critic networks $Q_{\psi_1}(s, \mathbf{a})$ and $Q_{\psi_2}(s, \mathbf{a})$ mitigate over-estimation bias through the minimum operator. In addition, target networks with soft updates τ ensure training stability, while the network architecture employs three hidden layers with 256-256-128 neurons and layer normalization for the high-dimensional state space.

C. Training Strategy

A curriculum learning accelerates convergence by progressively increasing the complexity in the scenario. Episodes 1–100 use mild scenarios (2 blocked areas, 3 survivor clusters); episodes 100–500 introduce moderate complexity (4 blocked areas, 5 clusters); subsequent episodes employ the full distribution including severe conditions (6 blocked areas, 7 clusters) with 50 dB attenuation in blocked zones. This progressive approach prevents early policy collapse in challenging scenarios whilst ensuring an eventual generalization. The experience replay with buffer size 10^5 and batch size 256 enables an efficient off-policy learning. An early stopping criterion with 500-episode patience terminates training when improvement falls below 0.1%.

IV. NUMERICAL RESULTS AND DISCUSSION

Without loss of generality, we consider a network with 7 UAVs each mounted with an IRS of 32 elements serving $K = 50$ ground users. The system is implemented using Python 3.8+ with Stable-Baselines3 for SAC and PyTorch for neural network training, and training proceeds over 2000 episodes with early stopping. Table II summarizes key simulation parameters.

TABLE II: SIMULATION PARAMETERS.

Parameter	Value	Parameter	Value
M	7	Learning rate	3×10^{-4}
N	32	γ	0.99
K	50	τ	0.005
Frequency	28 GHz	Batch size	256
B	10 MHz	Buffer size	10^5
κ_r	10 dB	Hidden layers	256-256-128
α	2.2	P_{hover}	130 W
σ^2	−90 dBm	v_{max}	20 m/s
h_{min}	50 m	h_{max}	200 m
P_{proc}	5 W	κ	$3.5 \text{ W} \cdot \text{s/m}$
η	0.01 W		

A. Algorithm Performance Comparison

Table III presents the performance comparison of seven ML algorithms evaluated over 50 episodes. DDPG achieves the highest reward of 0.815 with a minimal inference latency of 0.17 ms, making it suitable for deployment in static environments. However, SAC provides competitive performance with a reward of 0.649 and superior stability (standard deviation of 0.009 versus 0.002), which proves advantageous for online learning requiring continuous adaptation to novel scenarios. The PPO algorithm exhibits a higher variance of

TABLE III: PERFORMANCE COMPARISON OF ML ALGORITHMS.

Algorithm	Reward	Std Dev	Infer. (ms)	Stability
SAC	0.649	0.009	0.46	High
PPO	0.644	0.136	0.22	Medium
DDPG	0.815	0.002	0.17	High
DQN	0.819	0.004	0.35	High
Rand. Forest	0.585	0.110	5.22	Low
Gauss. Proc.	0.806	<0.001	0.20	High
LSTM	0.412	0.245	383.09	Very Low

0.136 due to constraints on the on-policy learning, requiring 50% more samples than the off-policy approach of SAC. Classical ML methods including Random Forest and Gaussian Process achieve moderate performance, but cannot handle a sequential decision-making inherent to trajectory optimization. Notably, long short-term memory (LSTM) suffers a prohibitive latency of 383 ms, rendering it impractical for real-time UAV control where decisions must occur within the 100-ms channel coherence window.

B. Energy Efficiency Performance

Fig. 2 illustrates the Pareto front comparing SAC with some traditional optimization methods. The proposed SAC framework achieves an improvement of 33% EE over baseline static configurations (73.9 Mbps/W versus 55.7 Mbps/W), with spectral efficiency gains of 9% (1810 bps/Hz versus 1658 bps/Hz). In addition, SCA reaches 1705 bps/Hz with 68.2 Mbps/W, whilst BCD achieves 1723 bps/Hz with 63.4 Mbps/W. The performance gap arises due to the fact that SCA and BCD algorithms converge to local optima through alternating convex decomposition, whereas SAC's entropy-regularized exploration navigates the non-convex 70-dimensional action space more effectively.

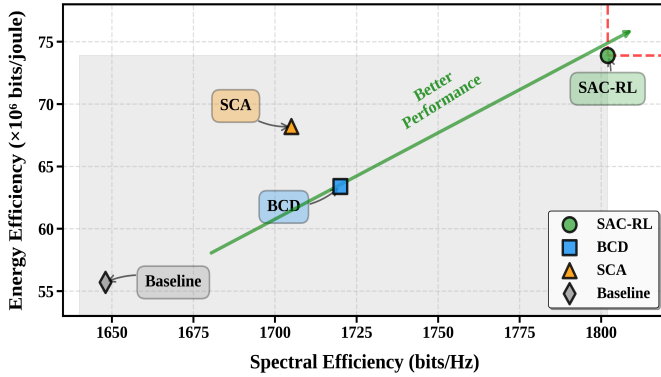


Fig. 2: Pareto front showing spectral efficiency versus energy efficiency trade-off.

From an average power consumption, the proposed SAC algorithm shows a decrease from 297.6 W to 232.1 W, translating into an extended mission duration from 3 hours to approximately 4.3 hours for a battery capacity of 1 kWh, i.e., an increase of 43% in operational time. Furthermore, the quantitative decomposition through ablation studies reveals that trajectory optimization contributes 35% of total gains (i.e., 11.5% absolute improvement), while the optimization of the IRS phase shift contributes 65% (i.e., 21.5% absolute improvement). This result appears counter-intuitive given that the

propulsion mechanism dominates UAV power consumption with more than 96%, yet trajectory optimization yields only 35% gains in efficiency. Table IV highlights the power breakdown confirming the propulsion dominance. The explanation lies in the multiplicative nature of the effective channel in (2): optimized phase shifts achieve constructive interference that provides 12–15 dB passive beamforming gain, reducing the required transmission power by approximately 40%. Subsequently, this improvement in channel quality enables trajectory planning that reduces unnecessary repositioning by 35%, as UAVs can maintain an acceptable quality of service (QoS) from fewer positions.

TABLE IV: BREAKDOWN IN POWER CONSUMPTION.

Component	Power (W)	Percentage
UAV Propulsion	287.9	96.6%
IRS Phase Shifting	5.0	1.7%
Processing Circuits	5.0	1.7%
Total	297.9	100%

The phase coherence metric, defined as the average cosine similarity between achieved and optimal phase configurations, reaches 0.82 compared to 0.21 for random configurations, confirming a strong constructive interference alignment. However, suboptimal phases cause a loss of 8–10 dB and a reduction in data rate of 40–50%. The synergistic coupling between trajectory and phase shift optimization yields an additional gain of 4 percentage points beyond the independent optimization. This synergy arises from optimized UAVs trajectories where phase alignment is most effective, while phase shifts adapt to trajectory-induced channel variations. When optimized separately using alternative methods as in [10], each component converges to locally optimal solutions that may not be globally coordinated. In addition, ablation experiments confirm that trajectory-only optimization achieves 22% improvement, phase-only achieves 27%, while joint optimization reaches 33% rather than the expected 49% sum, indicating that individual gains are not additive but exhibit diminishing returns when applied in isolation.

C. Computational Efficiency Analysis

Fig. 3 compares the execution time of the proposed SAC-based approach with the ones of existing ML algorithms.

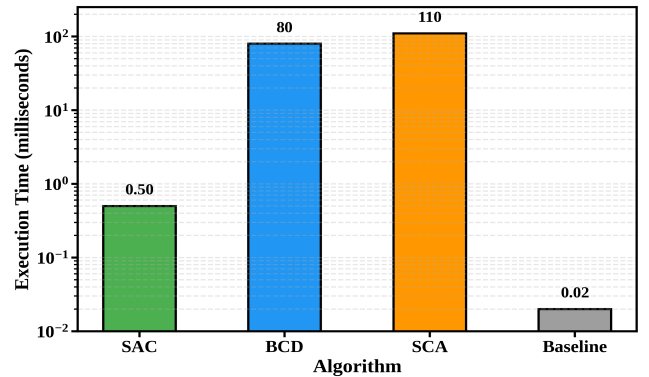


Fig. 3: Comparison of execution times for ML algorithms.

The SAC inference time of 0.46 ms enables real-time deployment, achieving 160 $\times$ speedup over BCD (80 ms)

and $220\times$ over SCA (110 ms). This speedup arises from fundamental algorithmic differences, i.e., traditional iterative methods require 10–30 iterations of $\mathcal{O}(N^3M)$ complexity per timestep to solve convex subproblems, whereas SAC performs single $\mathcal{O}(N^2)$ forward passes through trained networks. The training cost of the deployment is reduced with 2000 training episodes requiring approximately 4 hours on GPU, while subsequent inference incurs only the forward pass cost. The robustness analysis reveals that the system maintains 82% performance under 20% CSI estimation error, compared to 45% for BCD assuming perfect CSI. This robustness stems from SAC’s stochastic policy training, which exposes the network to action variations during learning and produces policies tolerant to input perturbations – an important characteristic given that IRS lacks radio frequency (RF) chains for direct channel measurement [9].

D. Emergent Behaviour and Coverage

An analysis of learned trajectories reveals emergent “hover-and-optimize” behaviour, where UAVs identify strategically advantageous positions and remain stationary while relying on phase optimization to track channel variations. This behaviour reduces propulsion energy (hovering requires 130 W versus up to 200 W during movement) while maintaining coverage through intelligent reflection. The learned policy positions UAVs at an average altitude of 127 m, balancing path loss against coverage area. The system maintains 95% user coverage compared to 80% with random phase configurations, with an average signal-to-noise ratio (SNR) of 28.5 dB. Moreover, battery management achieves zero depletion failures with 31% average remaining capacity.

E. Practical Design Implications

The results yield actionable guidelines for UAV-IRS system designers. *First*, given that phase optimization contributes 65% of EE gains, system budgets should prioritize high-resolution phase shifters (4-bit minimum) over propulsion efficiency improvements. The 12–15 dB passive gain from optimal phase alignment justifies the marginal increase in IRS control complexity. *Second*, the “hover-and-optimize” emergent behaviour suggests that mission planning should identify strategic hovering positions *a priori*, reducing the action space and accelerating convergence during deployment. *Third*, for disaster scenarios requiring rapid deployment, pre-trained SAC policies can be fine-tuned with 50–100 episodes (approximately 15 minutes) to adapt to site-specific channel conditions, compared to full retraining requirements of conventional methods.

V. CONCLUSION AND FUTURE WORK

This paper presented a SAC-based DRL framework for joint UAV trajectory and IRS phase shift optimization in a multi-UAV network. The proposed approach has been shown to achieve an EE improvement of 33%. Moreover, a quantitative decomposition revealed that optimization of the IRS phase shift contributes 65% of the total EE gain despite consuming less than 2% of the system power, while the optimization of the UAV trajectory provides the remaining 35%. Despite these promising results, several limitations still need to be

addressed. To begin with, the simulation setup assumes perfect UAV positioning with a global positioning system (GPS) accuracy of less than 1 m, whereas practical deployments experience errors of 3–5 m, resulting in the degradation of phase coherence. Moreover, the single-agent SAC requires centralized coordination. Specifically, multi-agent extensions using centralized training, decentralized execution (CTDE) may enable truly distributed operations. In addition, the overhead due channel estimation which consumes approximately 5% of coherence time, is not explicitly modeled leading to a reduction in the effective throughput. Finally, hardware validation remains essential, e.g., IRS prototypes exhibit 1–2 dB insertion loss not captured in simulations. Potential avenues for future work will address these limitations. Moreover, an interesting avenue will explore a multi-agent SAC scenario for large-scale urban deployments and hardware-in-the-loop validation.

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